

Cloudflare Radar 2024 Year in Review

Data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's network

17%

Global traffic growth

225

Major disruptions

6.5%

Traffic mitigated

Audience: Technology executives and network infrastructure leaders

Tone: professional, data-driven, authoritative

17% Global Traffic Growth and Starlink's 3.3x Surge

Traffic demand continued upward while satellite connectivity expanded reach into underserved markets.

17%

YoY global Internet traffic increase

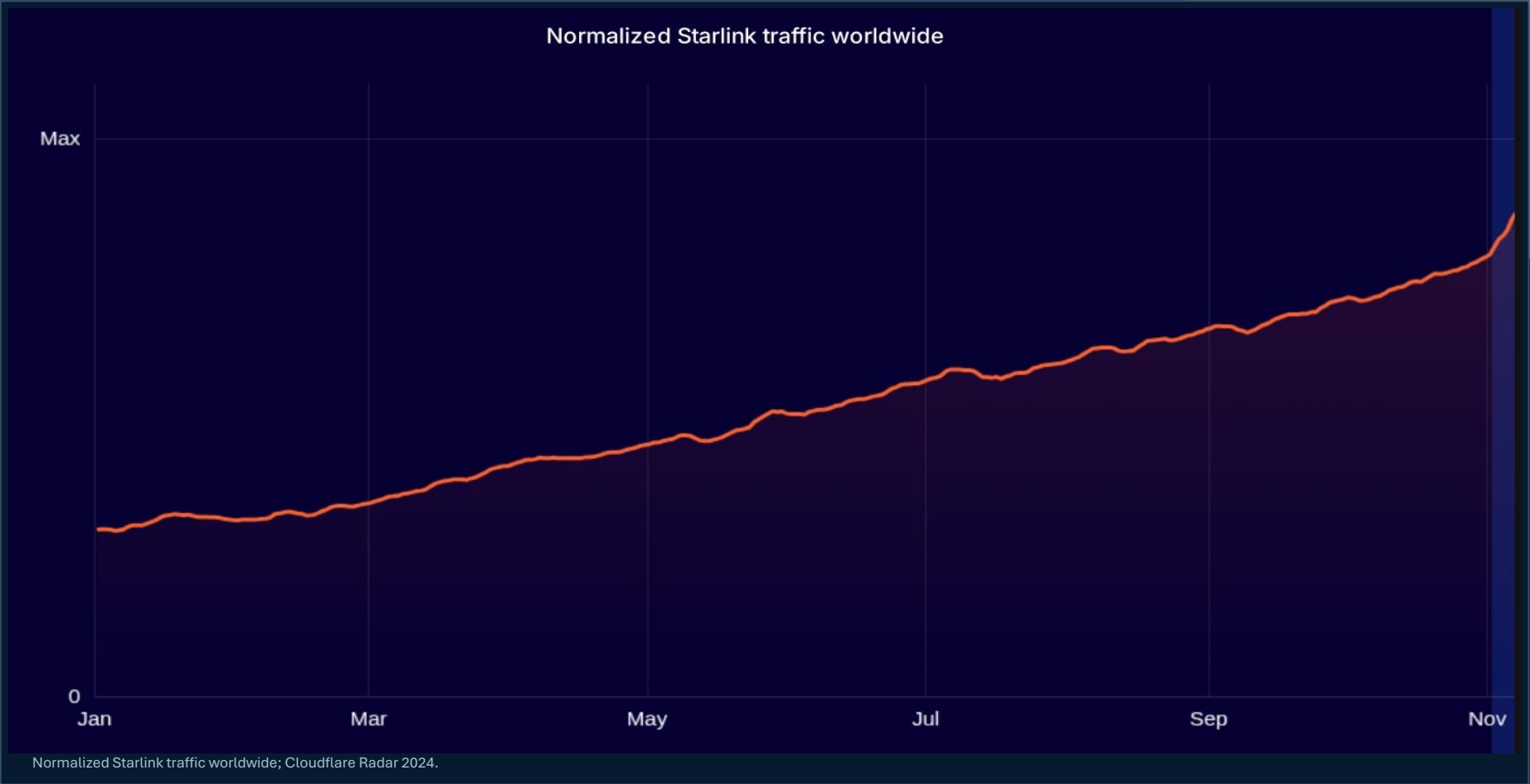
Trend normalized from mid-January post-holiday

3.3x

SpaceX Starlink traffic growth

Scaled relative to peak observed request volume

- Reliance on Internet services continues to compound across communication, commerce, entertainment, and transportation
- Starlink growth signals demand for resilient last-mile connectivity where terrestrial networks remain limited.



Executive implication: capacity planning must assume continued baseline traffic growth plus localized surges from new access networks.

Google Leads Top 10 Services as Generative AI Rises

DNS resolver data shows durable platform concentration while generative AI continues its rapid rise.

TOP 10 INTERNET SERVICES	
Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

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Top 10 most popular Internet services worldwide.

RANKING METHOD

Based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver.

MARKET SHIFT

Generative AI continued its rapid rise throughout 2024, while the Metaverse receded.

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Protocol, encryption, and developer ecosystem shifts accelerated across the global Internet stack.



2%

Early-2024 PQ share

→

13%

Approx. peak share

TRIGGER EVENT

- Chrome 124 enabled post-quantum key agreement by default on April 17.
- Mozilla Firefox began rollout by default; Apple Safari began initial testing.

21%

HTTP/3 traffic adoption

33%

Traffic from iOS devices

28%

IPv6 dual-stack traffic

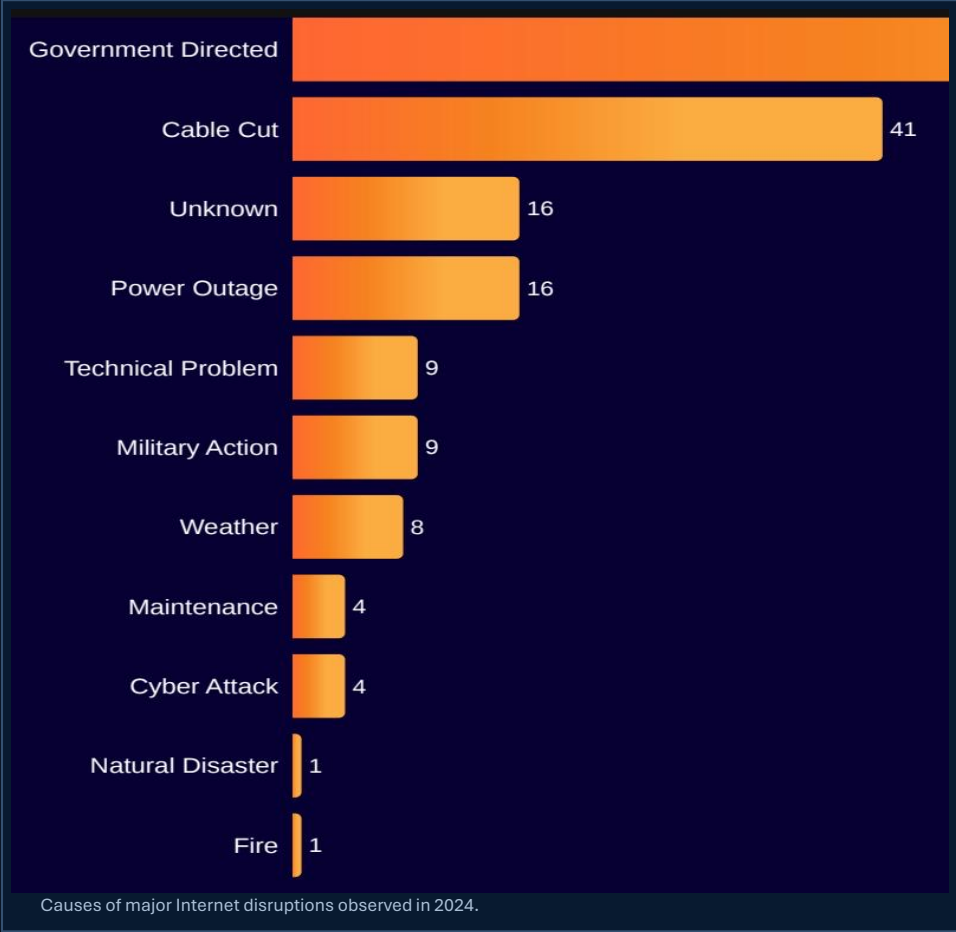
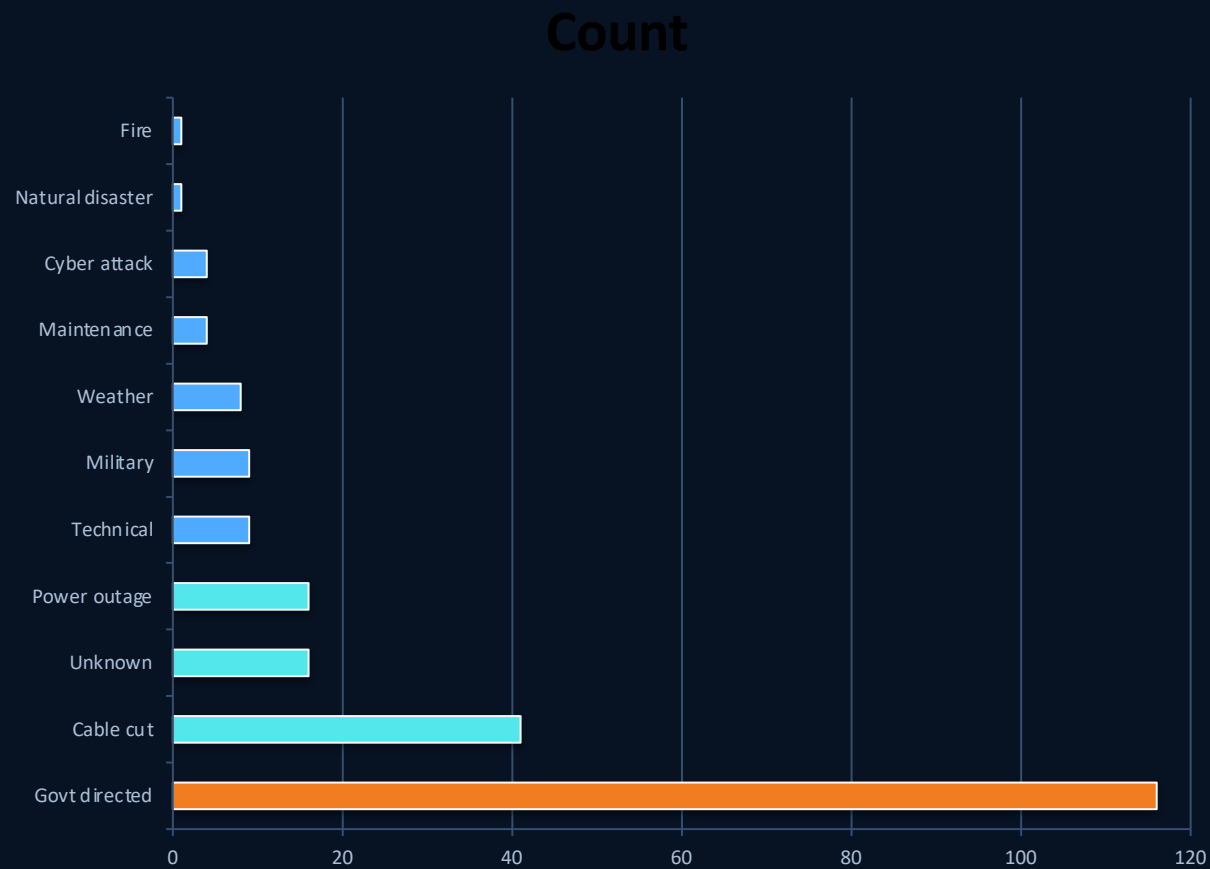
12%

Go API client share

Go became the top API client language, followed by Node.js, Python

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Cloudflare observed disruptions across policy, physical infrastructure, power, conflict, weather, and cyber causes.



Key point: government-directed shutdowns accounted for 116+ of 225 major disruptions, ahead of cable cuts (41) and power outages (16).

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Threat activity remained broad-based, automated, and persistent across web traffic, bots, vulnerabilities, and email.

6.5%

All processed traffic mitigated

68%

Bot traffic from top 10 countries

MOST-ATTACKED SECTOR

Gambling and Games was the most attacked industry category.

4.3%

Emails processed were malicious

43%

Malicious emails had deceptive links

PERSISTENT EXPLOITATION

Log4j (CVE-2021-44228) continued to be actively targeted more than two years after disclosure.

EMAIL THREAT METRICS

Malicious emails	4.3%
Malicious with deceptive link	43%
Top malicious/spam TLD	.bar

ROUTING SECURITY

RPKI valid routes increased by 6.4%, strengthening Internet routing security posture.

Five Takeaways Shaping the 2025 Internet Outlook

Strategic priorities for network, platform, and security leaders planning for 2025.

1

Plan for compounding demand

Global traffic grew 17% year-over-year; Starlink surged 3.3x as satellite access expanded.

2

Modernize cryptography roadmaps

Post-quantum encryption climbed from ~2% to 13% of TLS 1.3 traffic after Chrome 124.

3

Design for connectivity fragility

Government-directed shutdowns led the 225 major disruption events, followed by cable cuts and power outages.

4

Assume persistent threat pressure

6.5% of traffic was mitigated; Gambling/Games led attacked sectors and Log4j exploitation persisted.

5

Track developer ecosystem shifts

Go became the top API client language at roughly 12% of automated API requests.

2025 outlook: resilience, cryptographic agility, and threat-informed automation become board-level infrastructure priorities.